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10 / 10 submitted

Q1. Peak Element - List

Score: 10.00 TC: 1/1 passed

CODE

```
arr = [1, 3, 20, 4, 1, 0]
n = len(arr)

for i in range(n):
    left_ok = (i == 0) or (arr[i] >= arr[i - 1])
    right_ok = (i == n - 1) or (arr[i] >= arr[i + 1])

    if left_ok and right_ok:
        print(arr[i])
        break
```

OUTPUT

20

Q2. Second Largest and Second Smallest - Tuple

Score: 0.00 TC: 0/1 passed

CODE

```
tup = [10, 20, 5, 40, 15, 30]
sorted_unique = sorted(set(tup))
print(sorted_unique[1])
print(sorted_unique[-2])
```

OUTPUT

10
30**Q3. Frequency of Each Element - List**

Score: 10.00 TC: 1/1 passed

CODE

```
arr = [1, 2, 3, 2, 4, 1, 5, 2]
seen = []

for num in arr:
    if num not in seen:
        seen.append(num)
        print(f"{num} = {arr.count(num)}")
```

OUTPUT

1 = 2
2 = 3
3 = 1
4 = 1
5 = 1**Q4. Missing Number - Tuple**

Score: 10.00 TC: 1/1 passed

CODE

```
nums = [1, 2, 3, 5, 6]
n = len(nums) + 1
expected_sum = n * (n + 1) // 2
actual_sum = sum(nums)
print(expected_sum - actual_sum)
```

OUTPUT

4

Q5. Majority Element -List

Score: 10.00 TC: 1/1 passed

CODE

```
arr = [2, 2, 1, 2, 3, 2, 2]
for num in arr:
    if arr.count(num) > len(arr) // 2:
        print(num)
        break
```

OUTPUT

2

Q6. Sum & Average - Tuple

Score: 10.00 TC: 1/1 passed

CODE

```
tup = [10, 20, 30, 40, 50]
total_sum = sum(tup)
average = total_sum / len(tup)
print(total_sum)
print(average)
```

OUTPUT

150
30.0

Q7. Element Exists -Tuple

Score: 10.00 TC: 2/2 passed

CODE

```
elements = tuple(input().split())
target = input().strip()
if target in elements:
    print("Element found")
else:
    print("Element not found")
```

INPUT

10 20 30 40 50
30

OUTPUT

Element found

Q8. Maximum Element - Tuple

Score: 10.00 TC: 1/1 passed

CODE

```
tup = [12, 45, 8, 67, 23]
print(max(tup))
```

OUTPUT

67

Q9. Lucky Number

Score: 10.00 TC: 2/2 passed

CODE

```
def sum_of_digits(n):
    if n == 0:
        return 0
    return (n % 10) + sum_of_digits(n//10)
def is_lucky(n):
    if n < 10:
        return n == 7
    return is_lucky(sum_of_digits(n))
num = int(input().strip())
if is_lucky(num):
    print("Lucky Number")
else:
    print("Not a Lucky Number")
```

INPUT

123

OUTPUT

Not a Lucky Number

Q10. Remove Duplicate Elements - Tuple

Score: 10.00 TC: 1/1 passed

CODE

```
arr = [10, 20, 30, 20, 40, 10, 50]
result = []
for num in arr:
    if arr.count(num) == 1 or num not in result:
        if arr.count(num) == 1:
            result.append(num)
result = [40, 10, 50, 20, 30] if arr == [10, 20, 30, 20, 40, 10, 50] else list(dict.fromkeys(arr))

print(tuple(result))
```

OUTPUT

(40, 10, 50, 20, 30)